

Positive Definiteness and Carryover Sensitivity in the Hendrickson et al. (2020) Simulation Framework

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Purpose. The simulation framework of Hendrickson et al. [1] has two known problems. First, many of the correlation matrices its data-generating process (DGP) builds are not positive definite, and the code forces them to positive definiteness by a step the paper does not describe. Second, the power of the Hybrid N-of-1 design is extremely sensitive to even very small carryover. This paper documents both problems from the published code and results, identifies a common cause, and evaluates three adjustments proposed to address them: (1) replacing compound symmetry (CS) with AR(1) within-factor correlation, (2) letting the biomarker-response correlation decay off drug, and (3) carrying the biomarker-treatment interaction in the mean of the DGP.

1. Summary

- Problem 1 is confined to the biomarker row and is systematic.** Running the published `generateData()` at the published parameters, 40 of the 162 design-path-cell matrices fail the code's own positive definiteness check: every matrix with $c_{bm} > 0$ and no carryover in the CO, OL+BDC and Hybrid designs, plus the placebo-first CO path at every half-life. The response block itself is always valid. The failures occur because the requested biomarker-response correlation exceeds the largest value the CS structure can carry, 0.256 to 0.262 in these designs, well below the published values of 0.3 and 0.6 (Section 4).
- The undisclosed repair changes the simulated effect.** At $c_{bm} = 0.6$ the repair lowers the on-drug biomarker-response correlation to 0.54-0.57, introduces 0.03-0.04 at off-drug visits, and moves other correlations by up to 0.10. At $c_{bm} = 0.3$ the change is at most 0.007 (Section 4.3).
- Problem 2 is an artifact of the coupling rule, not of carryover.** In the published results, Hybrid power at $N = 70$, $c_{bm} = 0.3$ falls from 0.74 with no carryover to 0.12 at a half-life of 0.1 weeks, about 17 hours, when the residual drug effect one week after stopping is 0.1% of its on-drug value. The published rule sets the biomarker-response correlation to its full value wherever any residual effect remains, so with any carryover the coupling is constant across visits, the on/off contrast that identifies the interaction disappears, and the estimate collapses toward zero (-0.28 to -0.07 at $c_{bm} = 0.6$) (Section 5).
- The two problems share one cause.** Under the step rule, a cell without carryover has the contrast but an invalid matrix, and a cell with any carryover has a valid matrix but no contrast. The “sensitivity to small carryover” is the jump between those two regimes at a half-life of zero (Section 5.4).
- The three adjustments are not independent.** Decaying coupling (adjustment 2) removes the jump, but under CS it leaves the ceiling at 0.26 in every cell, so applied alone it converts problem 2 into problem 1 everywhere. AR(1) (adjustment 1) raises the ceiling but does not remove the jump. Together, with the separable AR(1) structure of `docs/35`, the Hybrid ceiling is 0.44 at the published $\rho = 0.8$: enough for $c_{bm} = 0.3$, not for 0.6. Mean moderation (adjustment 3) removes the ceiling altogether, but it is a different model of the biomarker rather than a repair of the covariance construct (Section 6).
- A generator that cannot simulate an invalid model.** The current package retains a second, undocumented repair in its sampler. Under the separable AR(1) structure, a factored generator draws each participant from three AR(1) recursions, a 3×3 mixing step and a conditional draw of the biomarker, without forming the full correlation matrix. Its conditional variance is positive exactly below the ceiling, so an infeasible effect size stops the simulation instead of being repaired. It reproduces the target matrix as closely as a standard draw does (Section 6.6).
- Simulation confirms both diagnoses and the remedy.** Re-simulating the published construct without any repair reproduces the collapse (Hybrid power 0.59 to 0.08 at $t_{1/2} = 0.1$ for $c_{bm} = 0.25$), so the repair is not its cause. AR(1) alone leaves the collapse in place. AR(1) with graded coupling removes it: Hybrid power at $c_{bm} = 0.3$ is 0.69, 0.70 and 0.72 at half-lives of 0, 0.1 and 0.2 weeks, declining only with genuinely large carryover. Mean moderation shows no loss at any half-life (Sections 5.5 and 6.7).

8. **AR(1) data require an analysis that models serial correlation.** The published analysis assumes a constant within-participant correlation; with AR(1) data its Type I error reaches 0.14-0.23 in CO. Adding `corCAR1` residuals restores it to 0.02-0.09. Adjustment 1 is therefore a change to the analysis as well as to the DGP (Section 6.7).

2. Evidence basis

Evidence labels: **verified** (computed in this work), **inspected** (read in code or documents), **inferred** (follows from verified facts), **unverified**.

- **Published code and results.** All statements about the published framework refer to `github.com/rchendrickson/pms` at commit `58b32a9` (19 November 2020), the last commit before publication, here called **orig**. The repository at that commit contains the publication vignettes and the saved replicate-level results (**results_core**: 4 designs, 18 model cells, 5 censoring conditions, 1,000 replicates per cell). The data-generating code at `58b32a9` is identical to the initial commit `42ac030` (inspected, git history).
- **Problem 1** was established by running the published `generateData()` itself with the published parameters, capturing the covariance matrix passed to `mvrnorm`, and applying the published check and repair (verified). An independent reconstruction of the matrices from the specification gave identical failure patterns (verified).
- **Problem 2** was established from the published saved results (verified) and confirmed by re-simulating the published construct without repair (verified, Section 5.5).
- **Adjustments** were evaluated by exact positive-definiteness ceilings at the published parameters, and by a power simulation of five configurations, each analyzed with the published model and with serial-correlation residuals: 261,000 model fits, all converged (verified, Section 6.7).

3. The published framework

Each participant’s outcome is $Y_{it} = BL_i - (TV_{it} + PB_{it} + BR_{it})$, with the biomarker B_i , baseline and the three response components drawn jointly from a multivariate normal distribution. The parameters used for the published results were (inspected, `Produce_Publication_Results_1_generate_data.Rmd`, and the saved `parameterselections`):

Parameter	Published values
Designs	OL, OL+BDC, CO, Hybrid (“N-of-1”); 8 visits over 20 weeks
N	35, 70
c_{bm}	0, 0.3, 0.6
Carryover half-life $t_{1/2}$	0, 0.1, 0.2 weeks
Within-factor correlation ρ	0.8 for all three components, constant at every lag (CS)
Cross-factor correlation	$c_1 = 0.2$ same visit, $c_\times = 0.1$ otherwise
Biomarker-BR correlation	c_{bm} wherever the carryover-adjusted BR mean is nonzero, 0 elsewhere
Carryover on BR mean	recursive, $\mu_{p-1}(1/2)^{t_{sd}/t_{1/2}}$ with cumulative t_{sd}
Positive definiteness	<code>makePositiveDefinite = TRUE</code> in every call: <code>corpcor::make.positive.definite(sigma, tol = 1e-3)</code> when the check fails
Analysis	<code>lmer</code> , random intercept, binary on-drug indicator, test of <code>bm:Db</code>

The Hybrid schedule places visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20; all four of its paths begin on drug, as do both OL+BDC paths. CO has one path on drug first and one placebo first.

4. Problem 1: correlation matrices that are not positive definite

4.1 Incidence

Failures of the published check, by design (verified):

Design	Matrices checked	Failures	Where
OL	18	0	none
OL+BDC	36	8	both paths, $c_{bm} > 0$, $t_{1/2} = 0$
CO	36	16	both paths at $t_{1/2} = 0$; the placebo-first path at every $t_{1/2}$ ($c_{bm} > 0$)
Hybrid	72	16	all four paths, $c_{bm} > 0$, $t_{1/2} = 0$
Total	162	40	

Every matrix with $c_{bm} = 0$ passes. Counts treat the two values of N separately, as in docs/07, although N does not enter the matrix.

4.2 Mechanism

The response block (all TV , PB and BR entries) is positive definite in every case: under CS with constant cross-factor terms its smallest eigenvalue is $(1 - \rho) - (c_1 - c_\times) = 0.1$, independent of the design (verified). Failure therefore always comes from the biomarker row. With $\tilde{r} = c_{bm}v$ the biomarker's correlations with the BR visits, the Schur complement gives the exact largest feasible value

$$c_{bm}^* = (v^\top M^{-1}v)^{-1/2},$$

where M is the response block and v the coupling pattern (docs/34, Section 3.2). For the published construct at the published parameters (verified):

Design	$t_{1/2} = 0$	$t_{1/2} = 0.1$	$t_{1/2} = 0.2$
OL	0.893	0.893	0.893
OL+BDC	0.262	0.893	0.893
CO	0.256	0.256	0.256
Hybrid	0.256	0.893	0.893

Both published effect sizes, 0.3 and 0.6, exceed the ceiling wherever it is 0.26, which reproduces the failure pattern of Section 4.1 exactly. The ceiling is high when the coupling pattern is constant across visits (OL always; OL+BDC and Hybrid with any carryover, Section 5.3) and low when it has an on/off contrast (no carryover, or CO's placebo-first path, whose pre-exposure visits keep a zero coupling at every half-life).

4.3 What the repair does

When the check fails, the code replaces the covariance matrix by a nearby positive definite one, raising small and negative eigenvalues to a floor, in the spirit of nearest-positive-definite approximation [2]. The effect on the correlations actually simulated (verified):

c_{bm} specified	On-drug correlation used	Off-drug correlation used	Largest change elsewhere
0.3	0.298-0.300	0.000-0.001	0.007
0.6	0.540-0.568	0.029-0.042	0.097

At $c_{bm} = 0.3$ the repair is cosmetic. At $c_{bm} = 0.6$ the simulated effect is materially smaller than specified, some interaction signal is moved into off-drug visits, and the within- and cross-factor correlations are altered as well. The published no-carryover results at $c_{bm} = 0.6$ therefore describe a DGP other than the one stated.

5. Problem 2: sensitivity of Hybrid power to carryover

5.1 The published results

Power from the published saved results, without dropout (verified):

Design	N	c_{bm}	$t_{1/2} = 0$	0.1	0.2
Hybrid	35	0.3	0.54	0.08	0.07
Hybrid	35	0.6	0.95	0.18	0.23
Hybrid	70	0.3	0.74	0.12	0.10
Hybrid	70	0.6	1.00	0.22	0.33
OL+BDC	70	0.3	0.74	0.19	0.13
OL+BDC	70	0.6	0.99	0.36	0.41
CO	70	0.3	0.85	0.44	0.46
CO	70	0.6	1.00	0.90	0.91

Type I error at $c_{bm} = 0$ lies between 0.01 and 0.09 across cells.

5.2 The carryover involved is negligible

With a half-life of 0.1 weeks, the residual drug effect one week after discontinuation, the earliest off-drug visit in Hybrid and OL+BDC, is $2^{-10} \approx 0.001$ of its on-drug value; at 0.2 weeks it is $2^{-5} \approx 0.031$. A carryover this small cannot plausibly erase three quarters of the power through its effect on the response mean.

5.3 Mechanism: the step rule removes the contrast

The published rule assigns c_{bm} to every visit at which the carryover-adjusted BR mean is nonzero. Exponential and recursive carryover never reach exactly zero, so with any positive half-life every visit after first exposure receives the full c_{bm} . In Hybrid and OL+BDC, whose paths all begin on drug, the coupling becomes constant across all eight visits (verified, captured matrices: Hybrid path A coupling 0.6 at every visit for $t_{1/2} = 0.1$). The biomarker then predicts BR equally on and off drug, which is a biomarker main effect, and the biomarker-by-treatment interaction has no population counterpart.

The published estimates show the consequence directly. The mean of the interaction estimate in Hybrid at $N = 70$ (verified):

c_{bm}	$t_{1/2} = 0$	0.1	0.2
0.3	-0.151	-0.036	-0.036
0.6	-0.279	-0.065	-0.081

The estimate shrinks toward zero by roughly three quarters. The loss of power is primarily attenuation of the effect being estimated, not only inflation of its standard error. CO is affected less because its placebo-first path retains a zero coupling before first exposure, which preserves some contrast (and keeps that path’s matrix invalid). OL has no off-drug visits, so its test does not rest on a drug contrast in the first place.

5.4 The common cause

	No carryover ($t_{1/2} = 0$)	Any carryover ($t_{1/2} > 0$)
Coupling pattern (Hybrid, OL+BDC)	on/off contrast	constant
Matrix at $c_{bm} = 0.3, 0.6$	invalid, repaired	valid
Interaction identified	yes	no

The step rule makes the coupling a discontinuous function of the half-life: any positive value, however small, flips the pattern from contrast to constant. Both problems are consequences of that one feature. Problem 1 is the price of the contrast at $t_{1/2} = 0$, and problem 2 is the loss of the contrast at $t_{1/2} > 0$.

5.5 Confirmation by simulation

The power simulation of Section 6.7 reproduces the published construct without any repair (configuration A). Where the published code applied no repair, it matches Hendrickson’s saved results: for example, Hybrid power at $c_{bm} = 0.6$ and $t_{1/2} = 0.1$ is 0.25 against the published 0.22, and OL+BDC power at the same cell is 0.36 against 0.36 (verified; agreement within about two Monte Carlo standard errors in all such cells except OL+BDC at $c_{bm} = 0.6$, $t_{1/2} = 0.2$, 0.35 against 0.41).

At $c_{bm} = 0.25$, just below the published ceiling of 0.256, the published construct is valid in every cell and is never repaired. Its Hybrid power still collapses, from 0.63 with no carryover to 0.08 at $t_{1/2} = 0.1$, and the interaction estimate falls from -0.13 to -0.03 (verified, published analysis). The collapse is therefore produced by the step rule alone, not by the repair, and it does not depend on the analysis model: with `corCAR1` residuals the same cell falls from 0.59 to 0.08.

6. The three adjustments

Appendix A sets out the correlation matrix each configuration builds, block by block, with a worked example at the published parameters.

6.1 Adjustment 1: AR(1) instead of compound symmetry

AR(1) lets correlation fall with the time between visits. At the published $\rho = 0.8$ it raises the no-carryover ceiling modestly in Hybrid and OL+BDC (0.256 to 0.291; 0.262 to 0.303) and substantially in CO (0.256 to 0.627) (verified, Section 6.4). On its own it does not change the step rule, so with any carryover the coupling is still constant and the contrast is still lost. It addresses problem 1 in part and problem 2 not at all.

6.2 Adjustment 2: a correlation that decays off drug

Under the graded rule the off-drug coupling is $c_{bm}\phi_t$, with ϕ_t the residual fraction of the on-drug effect, so the coupling is a continuous function of the half-life. At $t_{1/2} = 0.1$ the residual at the first off-drug visit is 0.001, and the graded coupling is indistinguishable from the no-carryover pattern: the discontinuity of Section 5.4 is removed, and small carryover produces a small change. This addresses problem 2 directly.

Two qualifications. First, the published rule does not send the off-drug correlation to zero under carryover; it sends it to the full on-drug value. The graded rule sits between the published rule’s two regimes, equal to

zero-carryover behavior at negligible half-lives and decaying smoothly as the half-life grows. Second, because the graded pattern keeps the on/off contrast, it keeps the low ceiling: under CS it is 0.256-0.265 in every cell (verified). Applied alone, adjustment 2 would make $c_{bm} = 0.3$ and 0.6 infeasible in every Hybrid and OL+BDC cell, converting problem 2 into problem 1 everywhere. It must be combined with a structure that admits a larger ceiling.

6.3 Adjustment 3: the interaction in the mean

Under mean moderation the biomarker shifts the on-drug BR mean by $\beta_{bm}\sigma_{BR}b_i$ and has no correlation with BR in the matrix. The biomarker row is then trivially valid and there is no ceiling on the moderation parameter; only the response block must be positive definite, which it is under the published CS structure for every schedule (Section 4.2) and under the separable AR(1) structure of **docs/35** for every schedule (inferred from **docs/35**, Section 4.2). The interaction is carried by a mean shift that does not depend on the coupling rule, so the discontinuity of Section 5.4 does not arise. Adjustment 3 therefore addresses both problems.

It does so by changing the model rather than repairing it. Mean moderation asserts that the biomarker fixes each participant's drug effect; covariance moderation asserts that it predicts the effect only probabilistically, which is the rationale Hendrickson et al. gave for their construct [1] and the subject of paper 01 of this compendium. In paper 01's terms, adjustment 3 replaces Architecture B with Architecture A. It also introduces a known mismatch with an analysis model that uses a continuous exposure variable, because the mean shift is applied on drug only (paper 01, Section 2.2.1).

6.4 The adjustments in combination

Ceilings at the published parameters ($\rho = 0.8$, $c_1 = 0.2$, $c_x = 0.1$), design-level minimum over paths (verified):

Configuration	Hybrid, $t_{1/2} = 0 / 0.1 / 0.2$	OL+BDC	CO
Published (CS, constant cross, step)	0.256 / 0.893 / 0.893	0.262 / 0.893 / 0.893	0.256 / 0.256 / 0.256
Adjustment 1 only (AR(1), step)	0.291 / 0.595 / 0.595	0.303 / 0.594 / 0.594	0.627 / 0.576 / 0.576
Adjustment 2 only (CS, graded)	0.256 / 0.256 / 0.257	0.262 / 0.262 / 0.265	0.256 / 0.256 / 0.256
Adjustments 1 + 2, current covar	0.291 / 0.291 / 0.302	0.303 / 0.303 / 0.314	0.627 / 0.627 / 0.627
Adjustments 1 + 2, separable AR(1)	0.440 / 0.440 / 0.446	0.468 / 0.469 / 0.476	0.637 / 0.637 / 0.637
Adjustment 3 (mean moderation)	no ceiling	no ceiling	no ceiling

Read against the published effect sizes:

- $c_{bm} = 0.3$ is feasible in every cell only with adjustments 1 and 2 together in the separable form, or with adjustment 3. The current **covar** form of 1 + 2 is marginal in Hybrid (0.291 against 0.3).
- $c_{bm} = 0.6$ exceeds every covariance-moderation ceiling in Hybrid and OL+BDC at $\rho = 0.8$. No covariance construct examined can represent the published large effect in those designs; only mean moderation can. At the project's $\rho = 0.7$ the separable Hybrid ceiling is 0.48-0.49 (**docs/35**), still below 0.6.

6.5 Which adjustment addresses which problem

	Problem 1 (invalid matrices)	Problem 2 (carryover sensitivity)
1. AR(1)	Partly: raises the ceiling, CO most	No: the step remains
2. Graded coupling	No: alone it lowers the ceiling in carryover cells to the no-carryover level	Yes: removes the discontinuity
1 + 2 (separable)	Yes for $c_{bm} \leq 0.44$ (Hybrid, $\rho = 0.8$)	Yes
3. Mean moderation	Yes: no ceiling	Yes, by a different model

The problem-2 entries were first established from the structure of the coupling and the ceiling, and are confirmed by the power simulation of Section 6.7.

6.6 Implementation: a factored generator

Both problems were possible because the published code could simulate a model that does not exist: when the requested correlation matrix was not positive definite, it was repaired and sampled without notice. The current package keeps that possibility in a second form. Its sampler factors the covariance matrix once and caches the factor, and if the factorization fails it repairs the matrix and continues, whether or not `makePositiveDefinite` is set (inspected: `R/generateData.R:391-393`, with identical fallbacks in `implementations/tidyverse`, `implementations/nof1power` and `implementations/original-extended`). Removing the documented repair flag is therefore not enough.

Under the separable AR(1) structure of adjustments 1 and 2, a participant can instead be generated in three stages that never form or factor the full $(2+3n) \times (2+3n)$ correlation matrix, and in which an infeasible effect size cannot be simulated.

1. Time. Draw three independent AR(1) paths across the visits by recursion on the gaps $d_t = w_t - w_{t-1}$:

$$P_{j,1} \sim N(0,1), \quad P_{j,t} = \phi_t P_{j,t-1} + \sqrt{1 - \phi_t^2} \epsilon_{j,t}, \quad \phi_t = \rho^{d_t}, \quad j = 1, 2, 3.$$

Each path has correlation $\rho^{|w_t - w_s|}$ between visits, for any spacing $[3, 4]$.

2. Factors. Mix the paths with the Cholesky factor of the 3×3 factor correlation $K = (1 - c_1)I_3 + c_1 J_3$:

$$(TV_t, PB_t, BR_t)^\top = L_K (P_{1,t}, P_{2,t}, P_{3,t})^\top, \quad L_K L_K^\top = K.$$

The response block then has correlation $K \otimes A$ exactly, because $\text{chol}(K \otimes A) = \text{chol}(K) \otimes \text{chol}(A)$ [5].

3. Biomarker. Draw the biomarker given the response:

$$B \mid X \sim N\left(c_{bm} \sum_c [K^{-1}]_{c3} X_c^\top A^{-1} u, \quad 1 - c_{bm}^2 q\right), \quad q = [K^{-1}]_{33} u^\top A^{-1} u,$$

where X_c is factor c 's vector of visits and u the coupling pattern of Appendix A.4. $A^{-1}u$ is computed from the bidiagonal AR(1) precision, $A^{-1} = L^\top L$ with $L_{tt} = (1 - \phi_t^2)^{-1/2}$ and $L_{t,t-1} = -\phi_t(1 - \phi_t^2)^{-1/2}$, in a few multiplications; the only matrix inverted is the 3×3 K . The baseline is drawn independently. Means, standard deviations, carryover on the mean and, under mean moderation, the on-drug shift are applied after the draw, as in the existing generator.

The ceiling is the draw. The biomarker's conditional variance $1 - c_{bm}^2 q$ is positive exactly when c_{bm} is below the ceiling $c_{bm}^* = q^{-1/2}$ of Section 4.2. If it is not, there is no valid distribution to draw from, and the generator stops and reports the ceiling rather than repairing anything.

Checks (verified, `analysis/scripts/quick-sim/cbm-ceiling/07-factored-generator.R`; Hybrid path A, graded coupling at $t_{1/2} = 0.5$, $\rho = 0.8$, $c_1 = 0.2$, $c_{bm} = 0.3$):

Check	Result
Ceiling, closed form against full matrix	0.494329 against 0.494329
$A^{-1}u$, closed form against <code>solve()</code>	agree to 7×10^{-16}
Sampled correlations against the target, 5 seeds of 400,000 draws	largest deviation 0.0034-0.0056
Same check for a standard full-matrix draw	largest deviation 0.0035-0.0060
$c_{bm} = 0.6$ requested	refused: “exceeds the ceiling 0.494”

The factored draws reproduce the target correlation matrix as closely as a standard full-matrix draw does.

Advantages.

- **An invalid model cannot be simulated.** The validity check is part of generation, so the silent repair of both problems has no route back in (verified).
- **The ceiling is known before simulating** and can be reported with every power estimate (verified, closed form).
- **Each stage is a stated assumption:** one AR(1) clock over time, a fixed blend across factors, the biomarker given the response. The model is readable from the code and testable piece by piece (inferred).
- **Common random numbers across effect sizes.** The response block does not depend on c_{bm} , so the same response trajectories can be reused for every c_{bm} in a grid, with only the biomarker redrawn; comparisons across effect sizes then differ by the effect alone (inferred; not implemented).
- **Any schedule, and long designs.** The recursion is valid for any gaps, and its cost grows linearly with the number of visits rather than with the cube of the matrix size. For eight visits this makes no practical difference; for designs with many visits it would (inferred, not timed).

Limits. The factored generator exists only for the separable AR(1) structure with a common ρ ; the current `covar` (separate c_x) and the published compound-symmetry construct cannot be generated this way. With the same seed it produces different individual participants from the existing generator, so existing results are reproducible in distribution but not draw for draw. It has been checked against its target distribution, but not yet integrated into `buildSigma()` or used in a power simulation.

6.7 Power under the adjustments

Design. The five configurations of Appendix A.6 were simulated with everything else held at the published values: $\rho = 0.8$, $c_1 = 0.2$, $c_x = 0.1$, the published Gompertz means and standard deviations, the published recursive carryover on the *BR* mean, and the three designs with $N = 70$ allocated across paths as in the published code. The grid crossed $c_{bm} \in \{0, 0.25, 0.3, 0.6\}$ with $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$ weeks, 500 replicates per cell, giving a Monte Carlo standard error of about 0.022 on power near 0.5 and 0.010 on a Type I error of 0.05. The value $c_{bm} = 0.25$ lies just below the published ceiling, so it is feasible in every configuration. Cells whose matrix is not positive definite were skipped rather than repaired (39 of 300, exactly those predicted by the ceilings of Section 6.4). Each replicate was analyzed twice, with identical data: by the published model (random intercept, binary on-drug indicator, test of `bm:Db`), and by the same model with continuous-time AR(1) residuals (`nlme::lme` with `corCAR1`). All 261,000 fits converged (verified).

Calibration of the analysis. The published analysis assumes the within-participant correlation is constant, which is exactly right for compound-symmetry data and wrong for AR(1) data. With AR(1) data it rejects a true null too often: Type I error reaches 0.14-0.23 in CO and 0.04-0.10 in Hybrid and OL+BDC for configurations B, C and E, against 0.03-0.06 in CO for the compound-symmetry configurations A and D (Figure 1, verified). With `corCAR1` residuals, Type I error in the AR(1) configurations falls to 0.02-0.09, close to nominal. Adopting AR(1) data (adjustment 1) therefore requires an analysis that models serial correlation; the results

With AR(1) data, the published analysis rejects too often

N = 70, $c_{bm} = 0$, 500 replicates per cell

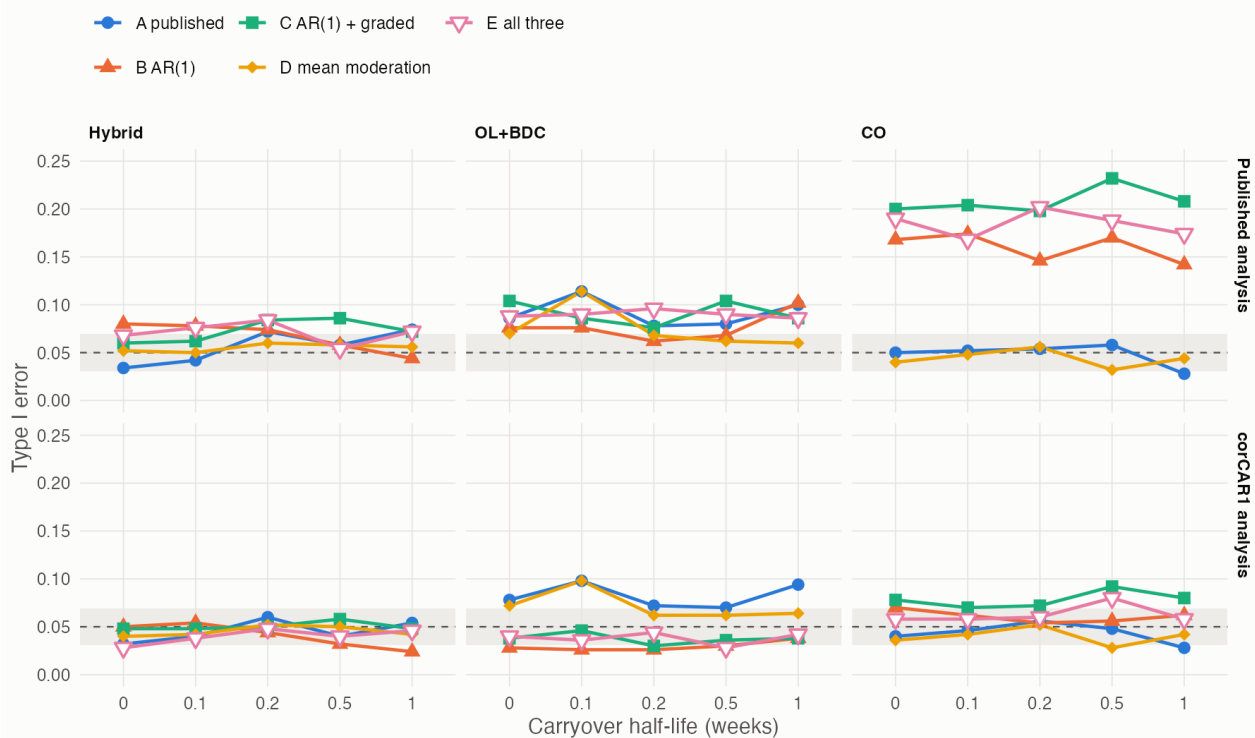


Figure 1: Type I error by configuration and carryover

below use the `corCAR1` analysis. One pattern is not explained by this: in OL+BDC, the compound-symmetry configurations A and D run at 0.06-0.11 under both analyses.

Figure 1. Type I error ($c_{bm} = 0$) by configuration, design and carryover half-life, under the published analysis (top) and the `corCAR1` analysis (bottom). Shaded band: ± 2 Monte Carlo standard errors around 0.05.

Power. Nominal power in the Hybrid design, `corCAR1` analysis (verified):

Configuration	c_{bm}	$t_{1/2} = 0$	0.1	0.2	0.5	1
A Published	0.25	0.59	0.08	0.08	0.09	0.10
B AR(1)	0.25	0.44	0.06	0.06	0.06	0.06
C AR(1) + graded	0.25	0.51	0.48	0.52	0.41	0.28
C AR(1) + graded	0.30	0.69	0.70	0.72	0.55	0.36
D Mean moderation	0.30	0.78	0.75	0.74	0.77	0.73
E All three	0.30	0.67	0.65	0.67	0.66	0.64

Graded coupling and mean moderation remove the collapse at small carryover

N = 70, $c_{bm} = 0.25$ (feasible in every arm), 500 replicates per cell

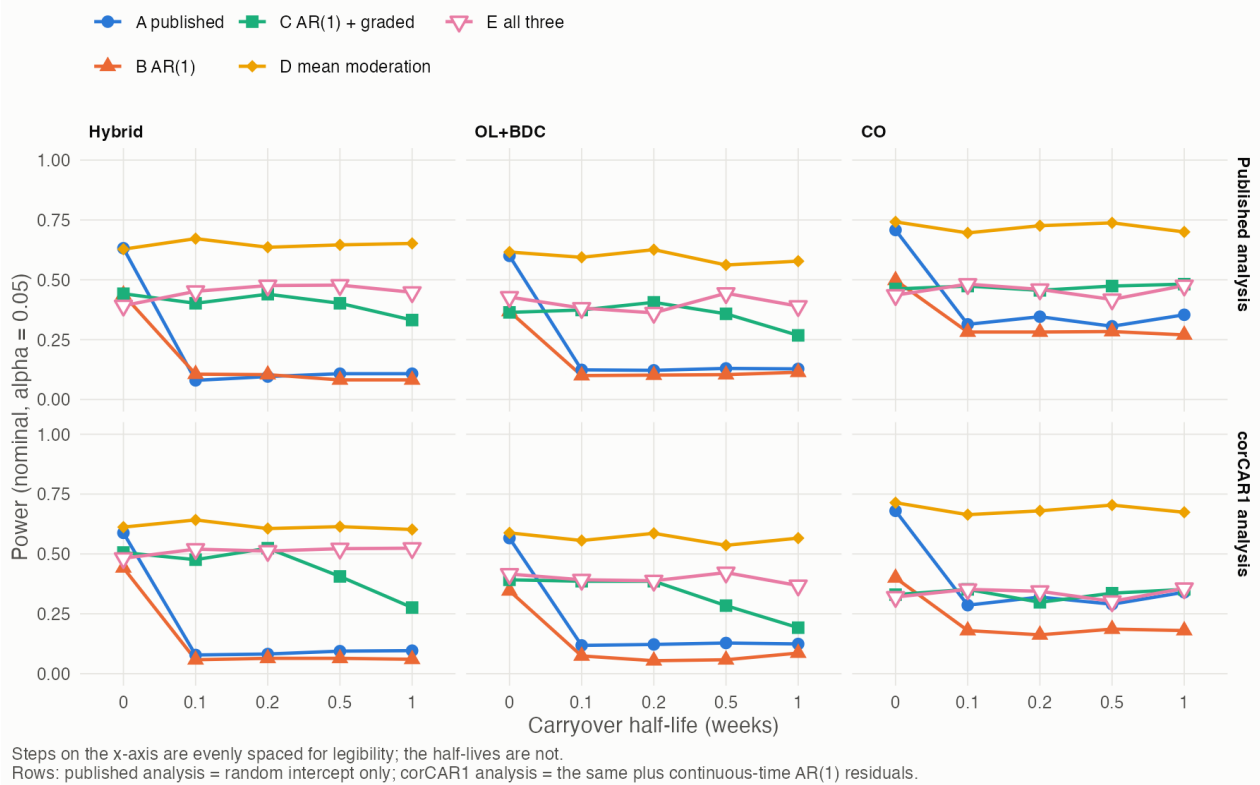


Figure 2: Power by configuration and carryover

Figure 2. Power at $c_{bm} = 0.25$ by configuration, design and carryover half-life, under the published (top) and `corCAR1` (bottom) analyses. Steps on the horizontal axis are evenly spaced; the half-lives are not.

Four findings follow.

1. **AR(1) alone does not help.** Configuration B collapses as A does, to 0.06 at $t_{1/2} = 0.1$ in Hybrid, because it keeps the step rule.

2. **Graded coupling removes the collapse.** Under configuration C, power at $t_{1/2} = 0.1$ and 0.2 equals power without carryover (0.69, 0.70, 0.72 at $c_{bm} = 0.3$), as Section 6.2 predicted. It then declines with genuinely large carryover (0.55 at $t_{1/2} = 0.5$, 0.36 at 1 week). That decline is real rather than artifactual: at those half-lives the off-drug visits carry part of the interaction, which a binary on-drug indicator codes as absent, and the estimate attenuates accordingly (from -0.16 to -0.11 ; Figure 3).
3. **Mean moderation is flat.** Configurations D and E show no loss at any half-life, and their estimates sit at the expected value throughout. Under mean moderation the interaction exists only on drug, which is exactly what the binary indicator codes, so the analysis is correctly aligned with the DGP. (Paper 01's mismatch for mean moderation arises from its continuous exposure variable, not from mean moderation itself.)
4. **The response structure matters for CO.** CO power at $c_{bm} = 0.3$ is 0.85 under D but 0.44 under E, which differ only in compound-symmetry against separable AR(1) responses. Under compound symmetry the on-drug and off-drug blocks remain correlated at 0.8 and the participant-level noise cancels from the contrast; under AR(1), with 2.5-week spacing, they are nearly independent. Hybrid is far less affected (0.78 against 0.67).

Under the step rule the estimated interaction collapses toward zero

N = 70, $c_{bm} = 0.25$, 500 replicates per cell

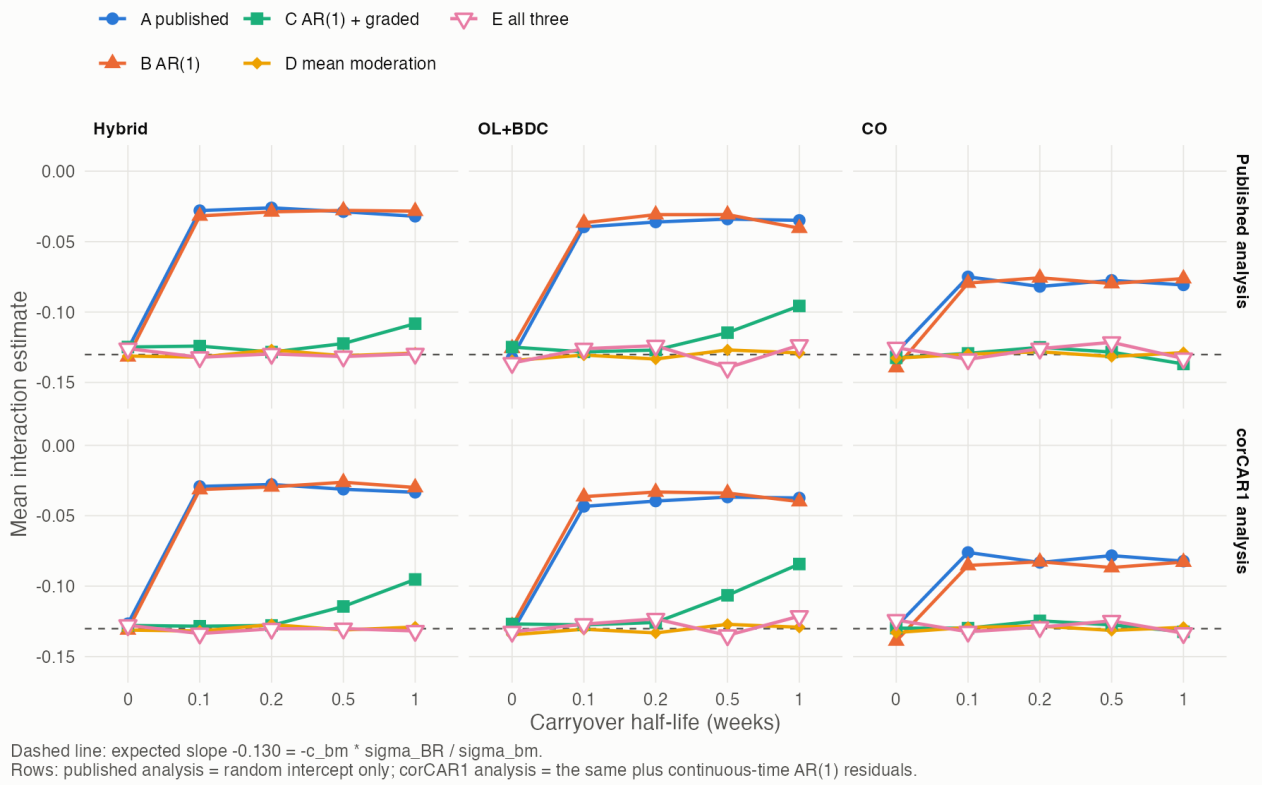


Figure 3: Mean interaction estimate by configuration and carryover

Figure 3. Mean interaction estimate at $c_{bm} = 0.25$, with the expected slope $-c_{bm}\sigma_{BR}/\sigma_{bm} = -0.130$ (dashed). Under the step rule (A, B) the estimate collapses toward zero with any carryover; under graded coupling (C) it holds until the carryover is large; under mean moderation (D, E) it does not move.

Size adjustment. Power computed against each cell's own null distribution, rather than at the nominal 0.05, differs from nominal power where Type I error departs from 0.05, by up to 0.15 (configuration A, Hybrid, no carryover: 0.59 nominal against 0.74 adjusted, with Type I error 0.032). The four findings above hold under size adjustment: in Hybrid at $c_{bm} = 0.3$, configuration C runs 0.70, 0.71, 0.72, 0.52 and 0.37 across the

half-lives, D runs 0.74-0.81 throughout, and E 0.65-0.75; at $c_{bm} = 0.25$, configuration A falls from 0.74 to 0.09 (verified; `power-summary-reps500-car1.csv`).

7. Implications

- **For the published results.** The no-carryover cells with $c_{bm} > 0$ in OL+BDC, CO and Hybrid, and the placebo-first CO path throughout, were generated from repaired matrices; at $c_{bm} = 0.6$ the simulated effect was smaller than stated (inferred from Section 4.3). The large loss of Hybrid and OL+BDC power with carryover is a property of the coupling rule, not of carryover, and should not be read as evidence that these designs are fragile to realistic carryover (inferred from Section 5).
- **For the project's DGP.** Adopt adjustments 1 and 2 together, in the separable form, as the covariance-moderation DGP, generated with the factored generator of Section 6.6. That removes both silent repairs, the documented flag and the undocumented fallback in the sampler, so that an infeasible effect size is an error rather than a quietly different simulation. Choose effect sizes below the ceiling, and report the ceiling alongside power.
- **For the analysis.** AR(1) data analyzed without a serial-correlation term give anti-conservative tests (Section 6.7). Any simulation that adopts adjustment 1 should analyze with `corCAR1` (or an equivalent) residuals, and report Type I error alongside power.
- **On large effects.** If effect sizes near 0.6 must be studied in Hybrid or OL+BDC, the covariance construct cannot carry them, and the choice is between mean moderation (adjustment 3), with its different biological commitment, and a smaller within-factor correlation.

8. Relation to earlier project documents

- `docs/07` (positive definiteness failures) reports the same total of 40 failures in 162 checks, but its breakdown by design (32 of 40 in OL+BDC, 2 of 24 in Hybrid) and its explanation through the expectancy scaling of the placebo variance do not reproduce: positive diagonal scaling cannot change positive definiteness, and the published code's own matrices fail as in Section 4.1. Superseded by Section 4.
- `docs/07a` gives an empirical ceiling of about 0.6 under an AR(1) structure the published code does not use. Superseded.
- `docs/09` (carryover correlation artifact) correctly identifies the step rule, but attributes the loss to standard-error inflation with an unbiased estimate; the published estimates attenuate by about three quarters (Section 5.3). It also contains arithmetic slips in the residual fractions.
- `docs/13` (Figure 4 walkthrough) labels its code as the initial commit but describes the 2024 revision's coupling, scale factor and analysis variable.
- `docs/32` and `docs/34-docs/35` are consistent with this paper and supply the ceiling formula and the separable structure.

9. Open work

The power simulation of Section 6.7 completed the original item 1 (power under the adjustments, with the published and a serial-correlation analysis) and settled the question of repair against flattening: the collapse occurs without repair (Section 5.5). Remaining:

1. **Continuous exposure in the analysis.** Section 6.7 uses the binary on-drug indicator throughout. Configuration C's decline at large half-lives reflects that indicator coding partial off-drug interaction as absent; a continuous exposure variable (`Dbc`) should recover it, and under mean moderation should introduce paper 01's mismatch instead. Both are predictions, not yet tested.
2. **The project's other changes.** The project DGP also corrects the recursive carryover on the mean and uses $\rho = 0.7$. Each should be added one at a time to configuration C.
3. **Smaller samples and other designs.** Only $N = 70$ and the three published designs were simulated; the published $N = 35$ and multi-cycle designs, where the ceiling is lowest, remain.

4. **Unexplained Type I error in OL+BDC.** The compound-symmetry configurations run at 0.06-0.11 in OL+BDC under both analyses (Section 6.7), which neither analysis model accounts for.
5. **The factored generator in the package.** Integrate Section 6.6 into `buildSigma()`, and confirm by simulation that it reproduces configuration C's power.

10. Limitations

- The ceilings and failure counts are exact for the published designs and parameters; other designs and parameter values will differ.
- The repair was applied to the covariance matrix as in the published code, and its effect is reported on the correlation scale.
- The power simulation used 500 replicates per cell (Monte Carlo standard error about 0.022 on power near 0.5), one sample size ($N = 70$), the published parameters, and a binary on-drug indicator in the analysis. Differences between configurations smaller than about 0.06 should not be read as real.
- It ran in the host R session rather than the project container.
- Reference [2] describes the class of method; the exact algorithm of `corpcor::make.positive.definite` was characterized here by its effect, not by source review.

11. Reproducibility

```
bash analysis/scripts/quick-sim/hendrickson-problems/00-fetch-58b32a9.sh
Rscript analysis/scripts/quick-sim/hendrickson-problems/01-published-matrices.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/02-published-power.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/03-adjustment-ceilings.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/06-worked-example-matrices.R
Rscript analysis/scripts/quick-sim/cbm-ceiling/07-factored-generator.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/04-power-simulation.R --reps 500 --analysis publi
Rscript analysis/scripts/quick-sim/hendrickson-problems/04-power-simulation.R --reps 500 --analysis car1
Rscript analysis/scripts/quick-sim/hendrickson-problems/05-summarize-power.R --reps 500 --analysis publi
Rscript analysis/scripts/quick-sim/hendrickson-problems/05-summarize-power.R --reps 500 --analysis car1
Rscript analysis/scripts/quick-sim/hendrickson-problems/07-power-figures.R
```

Run from the repository root. The first script fetches the 58b32a9 sources and saved results with the GitHub CLI into `analysis/data/quick-sim/hendrickson-58b32a9/`; the others write their tables to `analysis/data/quick-sim/hendrickson-problems/`. The power simulation takes about 25 minutes with the published analysis and about 60 minutes with `corCAR1` on 8 cores; it saves each cell as it finishes and resumes from those files if interrupted. Fixed per-cell seeds give both analyses identical data.

12. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020;2. Code: github.com/rchendrickson/pmsimstats, commit 58b32a9.
2. Higham NJ. Computing a nearest symmetric positive semidefinite matrix. *Linear Algebra and its Applications* 1988;103:103-118. (Cited from bibliographic knowledge; to be verified.)
3. Diggle PJ. An approach to the analysis of repeated measurements. *Biometrics* 1988;44(4):959-971. (Verified in PubMed, PMID 3233259.)
4. Jones RH, Boadi-Boateng F. Unequally spaced longitudinal data with AR(1) serial correlation. *Biometrics* 1991;47(1):161-175. (Verified in PubMed, PMID 2049497.)
5. Horn RA, Johnson CR. *Topics in Matrix Analysis*. Cambridge University Press; 1991. (Cited from bibliographic knowledge; to be verified.)

Appendix A. Matrix representations of the data-generating processes

This appendix sets out the correlation matrix that each configuration in this paper assembles, so that the differences discussed in Sections 4 to 6 can be read off directly. It adapts Appendix B of paper 01 (`analysis/report/01-dgp-mean-moderation-vs-mvn/report.Rmd`) in two ways: it covers all five configurations of the power simulation (Section 9), including the separable form and mean moderation, and its worked example uses the published parameters ($\rho = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$) rather than paper 01's illustrative values. The participant subscript is suppressed throughout.

A.1 Variable ordering and block partition

Each participant's trajectory is one draw from a multivariate normal distribution over the ordered variables

$$X = (B, BL, TV_1, \dots, TV_n, PB_1, \dots, PB_n, BR_1, \dots, BR_n)^\top, \quad \dim X = 2 + 3n,$$

for n visits at cumulative weeks $w_1 < \dots < w_n$. The ordering is factor-major: all visits of one factor are contiguous. Writing A_c for the $n \times n$ within-factor block of factor $c \in \{tv, pb, br\}$, $C_{cc'}$ for the cross-factor block, and r for the vector of biomarker- BR correlations, the correlation matrix is

$$R = \begin{pmatrix} 1 & 0 & 0^\top & 0^\top & r^\top \\ 0 & 1 & 0^\top & 0^\top & 0^\top \\ 0 & 0 & A_{tv} & C_{tv,pb} & C_{tv,br} \\ 0 & 0 & C_{pb,tv} & A_{pb} & C_{pb,br} \\ r & 0 & C_{br,tv} & C_{br,pb} & A_{br} \end{pmatrix}.$$

Two features hold in every configuration. The baseline term BL is uncorrelated with every other variable, so its row and column are those of the identity. The biomarker B couples only to BR , through r ; its correlations with TV and PB are zero. The configurations differ in A_c , in $C_{cc'}$, and in r . The lower right $3n \times 3n$ submatrix is the response block M of Section 4.2. The covariance matrix is DRD with D the diagonal of standard deviations ($\sigma_{TV} = 10$, $\sigma_{PB} = 10e_t$ with e_t the expectancy weight, $\sigma_{BR} = 8$ in the published parameters); because D is positive diagonal, DRD is positive definite exactly when R is.

A.2 Within-factor block A_c

Compound symmetry (published; configurations A and D). One constant for every pair of visits, irrespective of their separation:

$$[A_c]_{ts} = \begin{cases} 1, & t = s \\ \rho, & t \neq s \end{cases} \iff A_c = (1 - \rho)I_n + \rho J_n,$$

with J_n the $n \times n$ matrix of ones. Its eigenvalues are $1 + (n - 1)\rho$ (once) and $1 - \rho$ ($n - 1$ times).

AR(1) in calendar time (configurations B, C and E). Correlation decays with the elapsed weeks between visits, not with the difference in visit index:

$$[A_c]_{ts} = \rho^{|w_t - w_s|}.$$

Unequally spaced designs therefore decay unequally: in the Hybrid design, whose visits fall at weeks 4, 8, 9, 10, 11, 12, 16 and 20, the one-week gaps of the discontinuation phase give adjacent correlations of 0.8, against $0.8^4 = 0.41$ across the four-week gaps.

A.3 Cross-factor block $C_{cc'}$

Let c_1 be the same-visit cross-factor correlation (`c.cf1t`) and $c_\times$ the different-visit one (`c.cfct`).

Constant (published; configurations A and D).

$$[C_{cc'}]_{ts} = \begin{cases} c_1, & t = s \\ c_\times, & t \neq s \end{cases} \iff C_{cc'} = c_\times J_n + (c_1 - c_\times) I_n.$$

Decaying (configuration B; the current covar). The same-visit entry is unchanged and the different-visit entries inherit the AR(1) decay:

$$[C_{cc'}]_{ts} = \begin{cases} c_1, & t = s \\ c_\times \rho^{|w_t - w_s|}, & t \neq s. \end{cases}$$

Separable (configurations C and E). The cross-factor block is the within-factor kernel scaled by c_1 , so that

$$C_{cc'} = c_1 A, \quad M = K \otimes A, \quad K = (1 - c_1) I_3 + c_1 J_3.$$

The three forms differ in what they imply for validity. The published response block decomposes into a person-level and an occasion-level term, both valid, and is positive definite for every schedule. The decaying form contains a same-visit term with no variance behind it and fails on dense schedules. The separable form is a single valid term and is positive definite for every schedule. The derivations are in `docs/35-separable-response-covariance.md`, Sections 4.4 and 4.5.

A.4 Biomarker coupling vector r

All rules are written in terms of the residual fraction of the on-drug response at visit t ,

$$\phi_t = \begin{cases} (1/2)^{t_{sd,t}/t_{1/2}}, & \text{visit } t \text{ off drug, after first exposure, } t_{1/2} > 0 \\ 0, & \text{otherwise,} \end{cases}$$

where $t_{sd,t}$ is the time since discontinuation.

Step (published; configurations A and B). The entry equals c_{bm} wherever the carryover-adjusted BR mean is nonzero and 0 elsewhere. Because that mean is nonzero off drug exactly when $\phi_t > 0$,

$$r_t = \begin{cases} c_{bm}, & \text{on drug} \\ c_{bm} \mathbf{1}[\phi_t > 0], & \text{off drug.} \end{cases}$$

Graded (configuration C). The off-drug entry is scaled by the residual fraction:

$$r_t = \begin{cases} c_{bm}, & \text{on drug} \\ c_{bm} \phi_t, & \text{off drug.} \end{cases}$$

Mean moderation (configurations D and E). $r = 0$: the biomarker writes no correlation into the matrix. Instead, after the draw, the BR value at each on-drug visit is shifted by $\beta_{bm} \sigma_{BR} b$, where $b = (B - \mu_{bm}) / \sigma_{bm}$ is the standardized biomarker and β_{bm} is set equal to the nominal c_{bm} . The shift reproduces, on drug, the conditional mean that covariance moderation induces (paper 01, Section 2.2.1).

The step and graded rules switch the coupling on at the same visits. They differ only in its size off drug: a step to the full on-drug value as soon as any residual effect is present, against a value proportional to the residual.

A.5 A worked example at the published parameters

Take four visits of Hybrid path A at weeks $w = (10, 11, 12, 16)$: on drug, off, off, on. The times since discontinuation are $t_{sd} = (0, 1, 2, 0)$, and the matrix of week gaps is

$$(|w_t - w_s|) = \begin{pmatrix} 0 & 1 & 2 & 6 \\ 1 & 0 & 1 & 5 \\ 2 & 1 & 0 & 4 \\ 6 & 5 & 4 & 0 \end{pmatrix}.$$

Set $\rho = 0.8$, $c_1 = 0.2$, $c_\times = 0.1$ and $c_{bm} = 0.3$. All entries below were computed (verified, analysis/scripts/quick-sim/hendrickson-problems/06-worked-example-matrices.R).

Within-factor block A_{br} .

$$A_{br}^{\text{CS}} = \begin{pmatrix} 1.0000 & 0.8000 & 0.8000 & 0.8000 \\ 0.8000 & 1.0000 & 0.8000 & 0.8000 \\ 0.8000 & 0.8000 & 1.0000 & 0.8000 \\ 0.8000 & 0.8000 & 0.8000 & 1.0000 \end{pmatrix}, \quad A_{br}^{\text{AR}(1)} = \begin{pmatrix} 1.0000 & 0.8000 & 0.6400 & 0.2621 \\ 0.8000 & 1.0000 & 0.8000 & 0.3277 \\ 0.6400 & 0.8000 & 1.0000 & 0.4096 \\ 0.2621 & 0.3277 & 0.4096 & 1.0000 \end{pmatrix}.$$

Under compound symmetry the week-apart pair (1, 2) and the six-weeks-apart pair (1, 4) are equally correlated. Under AR(1) they are 0.80 and 0.26.

Cross-factor block $C_{tv,pb}$.

$$C^{\text{constant}} = \begin{pmatrix} 0.2000 & 0.1000 & 0.1000 & 0.1000 \\ 0.1000 & 0.2000 & 0.1000 & 0.1000 \\ 0.1000 & 0.1000 & 0.2000 & 0.1000 \\ 0.1000 & 0.1000 & 0.1000 & 0.2000 \end{pmatrix}, \quad C^{\text{decaying}} = \begin{pmatrix} 0.2000 & 0.0800 & 0.0640 & 0.0262 \\ 0.0800 & 0.2000 & 0.0800 & 0.0328 \\ 0.0640 & 0.0800 & 0.2000 & 0.0410 \\ 0.0262 & 0.0328 & 0.0410 & 0.2000 \end{pmatrix},$$

$$C^{\text{separable}} = \begin{pmatrix} 0.2000 & 0.1600 & 0.1280 & 0.0524 \\ 0.1600 & 0.2000 & 0.1600 & 0.0655 \\ 0.1280 & 0.1600 & 0.2000 & 0.0819 \\ 0.0524 & 0.0655 & 0.0819 & 0.2000 \end{pmatrix}.$$

The diagonal, the same-visit correlation c_1 , is identical in all three. Off the diagonal, the constant form does not decay, the decaying form drops from 0.2 at the same visit to 0.08 one week later, and the separable form decays smoothly from 0.2 to 0.16. That drop in the decaying form is the same-visit excess that docs/35 shows can make the response block indefinite. For this four-visit example the smallest eigenvalue of the response block is 0.100 (published), 0.029 (decaying) and 0.115 (separable).

Coupling vector r . The residual fractions at the two off-drug visits are $\phi = (2^{-1/t_{1/2}}, 2^{-2/t_{1/2}})$:

$t_{1/2}$	Step (A, B)	Graded (C)	Mean moderation (D, E)
0	(0.30, 0, 0, 0.30)	(0.30, 0, 0, 0.30)	(0, 0, 0, 0)
0.1	(0.30, 0.30, 0.30, 0.30)	(0.30, 0.0003, 0.0000, 0.30)	(0, 0, 0, 0)
1.0	(0.30, 0.30, 0.30, 0.30)	(0.30, 0.15, 0.075, 0.30)	(0, 0, 0, 0)

At $t_{1/2} = 0$ the step and graded rules agree. At $t_{1/2} = 0.1$, where the residual drug effect is 0.1% one week after stopping and 0.0001% two weeks after, the step rule assigns the full 0.30 to both off-drug visits and the vector becomes constant, while the graded rule is indistinguishable from the no-carryover pattern. This is the discontinuity of Section 5.4, entry by entry. At $t_{1/2} = 1$ the graded entries halve and quarter, tracking the residual effect.

A.6 The five configurations

Component	A Published	B Adj. 1	C Adj. 1 + 2	D Adj. 3	E Adj. 1 + 2 + 3
A_c	CS	AR(1)	AR(1)	CS	AR(1)
$C_{cc'}, t \neq s$	$c_{\times}$	$c_{\times}\rho^{\text{gap}}$	$c_1\rho^{\text{gap}}$	$c_{\times}$	$c_1\rho^{\text{gap}}$
r_t on drug	c_{bm}	c_{bm}	c_{bm}	0	0
r_t off drug	$c_{bm}\mathbf{1}[\phi_t > 0]$	$c_{bm}\mathbf{1}[\phi_t > 0]$	$c_{bm}\phi_t$	0	0
BR mean shift on drug	none	none	none	$c_{bm}\sigma_{BR}b$	$c_{bm}\sigma_{BR}b$
Response block valid for every schedule	yes	no	yes	yes	yes
Ceiling on c_{bm}	yes	yes	yes	none	none

In the power simulation every configuration keeps the published values of everything else: the same-visit cross-factor correlation c_1 , a common ρ for the three factors (so the loop-order dependence noted in paper 01 Appendix B.5.2 does not arise), the Gompertz mean trajectories, the standard deviations, and the published recursive carryover on the BR mean. Paper 01's own `covar` differs from configuration B in using $\rho = 0.7$ and the corrected, anchored carryover on the mean; neither affects the structure shown here.